A General-Purpose Algorithm for Life Table Construction

Integrating Relational Modeling, Smoothing, and Uncertainty

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A Disconnect between Training and Practice

Methods Training:

- Standard demographic texts treat mortality modeling and life table construction as separate exercises.
- Rates are treated as “given” before the life table is built.

Practice:

- National agencies regularly use ad-hoc combinations of models (B-splines, Logistic, Brass) for graduation and extrapolation, rarely reporting uncertainty.

What We Propose:

- Outline a general-purpose algorithm to combine mortality modeling, life table estimation, and construction of uncertainty intervals for any quantity of interest.

The Data Generating Process

Generalized linear model for death counts counts using a **Negative Binomial** likelihood to allow for overdispersion (θ):

$$_nD_x \sim \text{Negative Binomial}(_n\lambda_x, \theta)$$

We use a log-link with population exposure as an offset:

$$\log(_n\lambda_x) = \log(\mu(a)) + \log(_nE_x)$$

$$\log(\mu(a)) = X\phi$$

The Mean Value Theorem at Work

How do we map an age interval $[x, x+n)$ to an instantaneous rate $\mu(a^*)$?

The Mean Value Theorem for Integration guarantees an exact age a^* where the instantaneous hazard matches the interval rate:

$$\mu(a^*) = {}_nM_x$$

Rule of thumb:

- In the first year of life, mortality declines rapidly so $a^* < 0.5$. Use Andreev Kingkade (2015) procedure for ${}_1a_0$.
- $a^* = x + 0.5n$ for all other closed intervals.
- $a^* = 1/{}_nM_x$ for the open-ended interval.

A Brief Overview of Mortality Modeling

Before proposing a new model, we must understand the structural trade-offs of the classics:

- Gompertz (1825): Simple and effective log-linear model, but not applicable at young and old ages.
- Brass (1971): Relational model, appropriate for all ages but rigid if the target population has a different age trajectory.
- P-Splines (2000s): Purely local and flexible, but risk overfitting, (may) have poor extrapolation, and fail with sparse/low-quality data.

Can we combine the structural stability of Brass with the flexibility of splines?

Enter TOPALS: The Relational-Splines Bridge

The TOPALS approach (de Beer, 2012) bridges the gap between Brass and splines by modeling deviations from a standard schedule using linear splines.

- It was refined by Gonzaga and Schmertmann (2016), who replaced linear splines with B-splines and moved to a GLM framework
- Both versions require manual selection of the number and exact location of the knots (e.g., ages 0, 10, 20), requiring ad-hoc demographic judgment.
- This step is no longer required with modern P-splines.

The GARM Framework

We propose a **Generalized Additive Relational Model (GARM)** which extends TOPALS by using a more modern GAM estimation framework:

$$\log(\mu_a) = X\phi = \beta_0 + \beta_1 \log(m_a^{std}) + s(a)$$

$$s(a) = \sum_{j=1}^{k-1} \gamma_j b_j(a)$$

- $\log(m_a^{std})$: The standard schedule (e.g., HMD reference).
- $s(a)$: A penalized thin-plate regression spline.

Shrinkage: If the standard schedule perfectly explains the data, the penalty matrix shrinks $s(a)$ to zero.

Quantifying Uncertainty

In the GARM framework, we have that:

$$\phi \sim MVN(\hat{\phi}, \hat{V}_p)$$

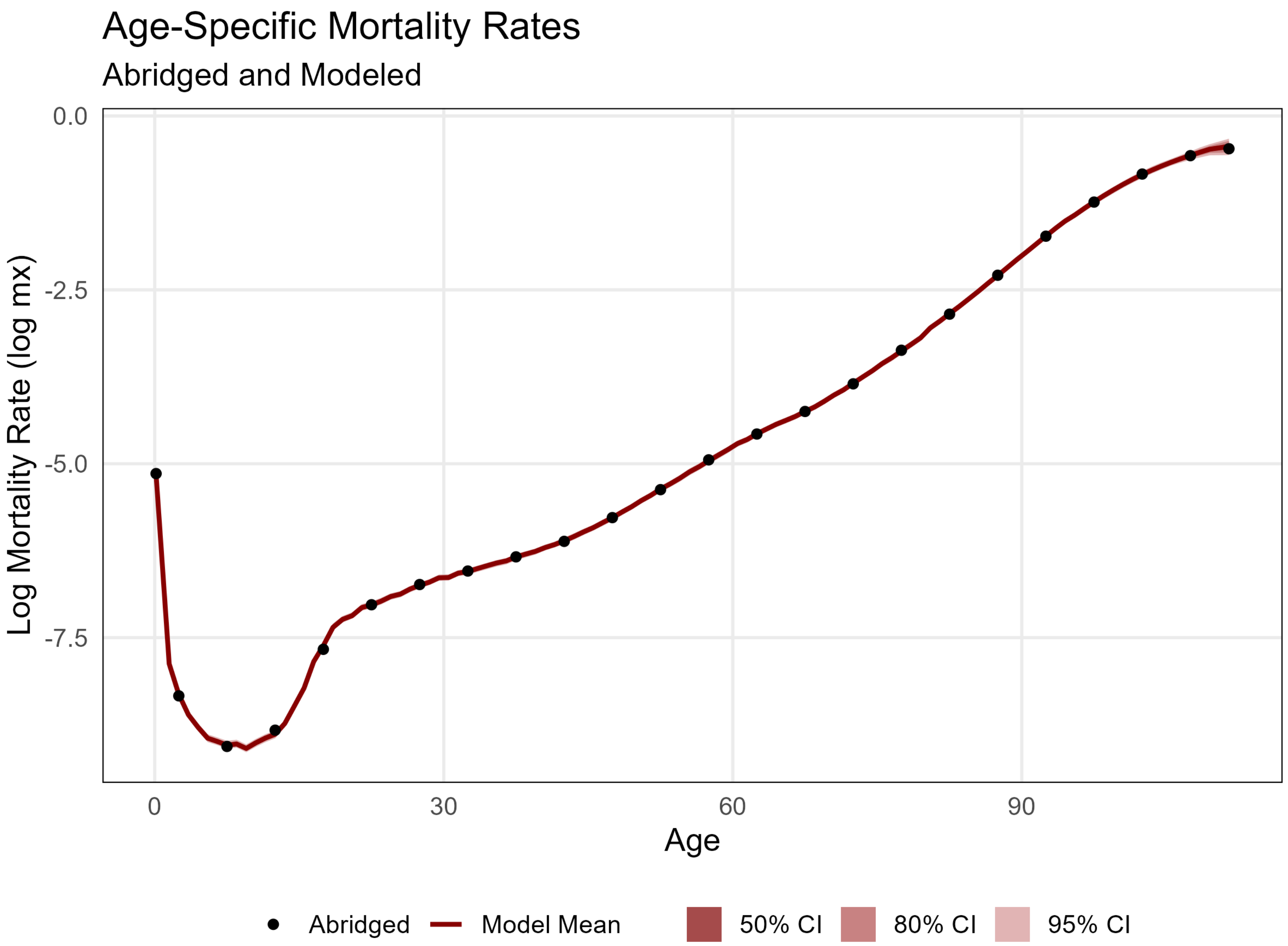
Because the coefficients are the only source of uncertainty for the log rates, this is all we need to simulate from the distribution of the log rates.

We can then propagate the uncertainty to any quantity that is a function of the log rates (such as life expectancy), by applying the required transformation to each set of simulated rates.

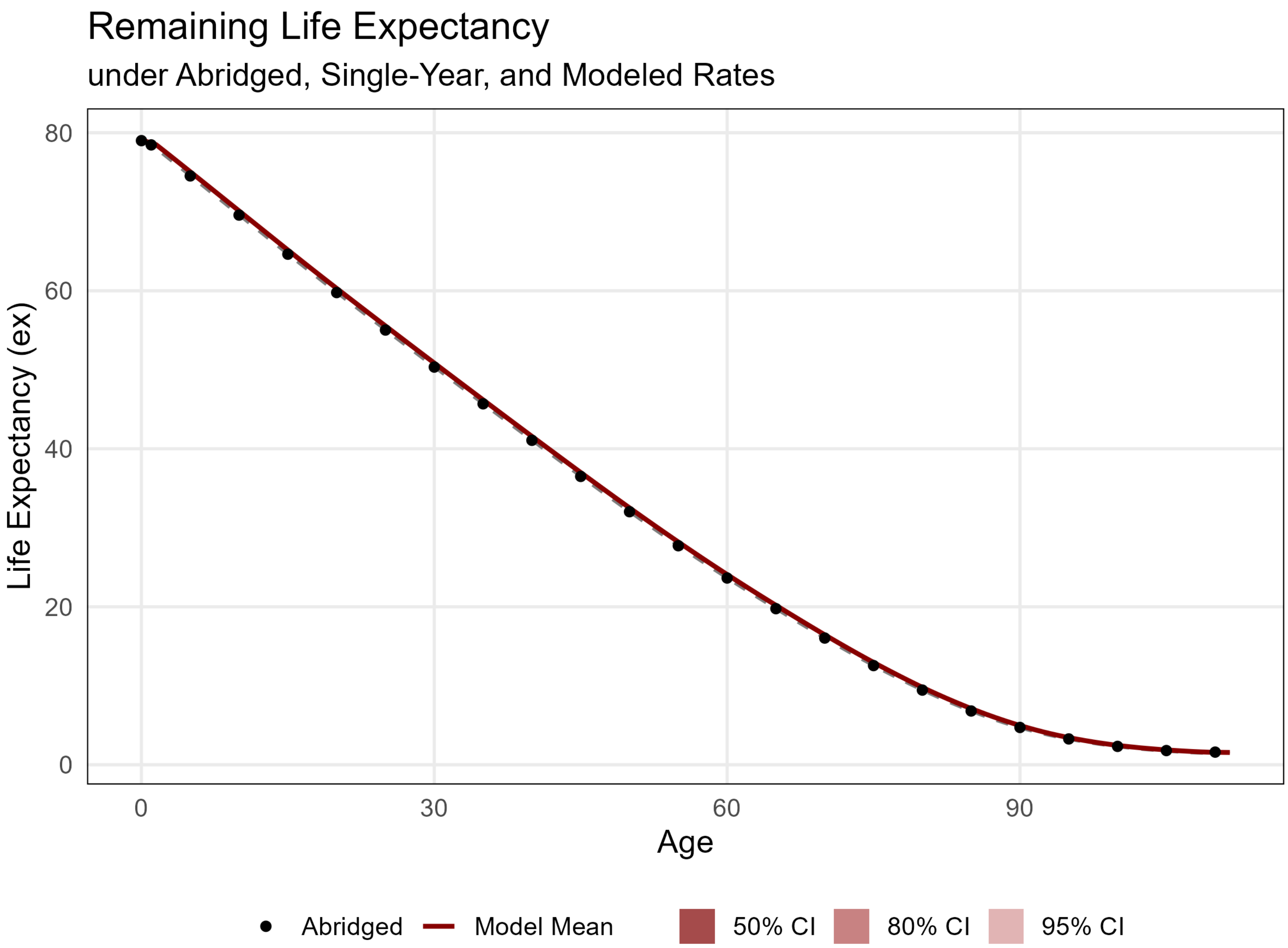
Scenario 1: U.S. 2019 Baseline Performance

We test the model on abridged U.S. 2019 data (HMD) to verify it can recover single-year rates without introducing bias.

For this and the next example we use 2015-19 data for the U.S. total population from HMD as the standard schedule.



Age-Specific Mortality Rates
Abridged and Modeled
0.0
-2.5
-5.0
-7.5
Log Mortality Rate (log mx)
0
30
60
90
Age
Abridged
Model Mean
50% CI
80% CI
95% CI



Remaining Life Expectancy
under Abridged, Single-Year, and Modeled Rates
Life Expectancy (ex)
80
60
40
20
0
0
30
60
90
Age
Abridged
Model Mean
50% CI
80% CI
95% CI

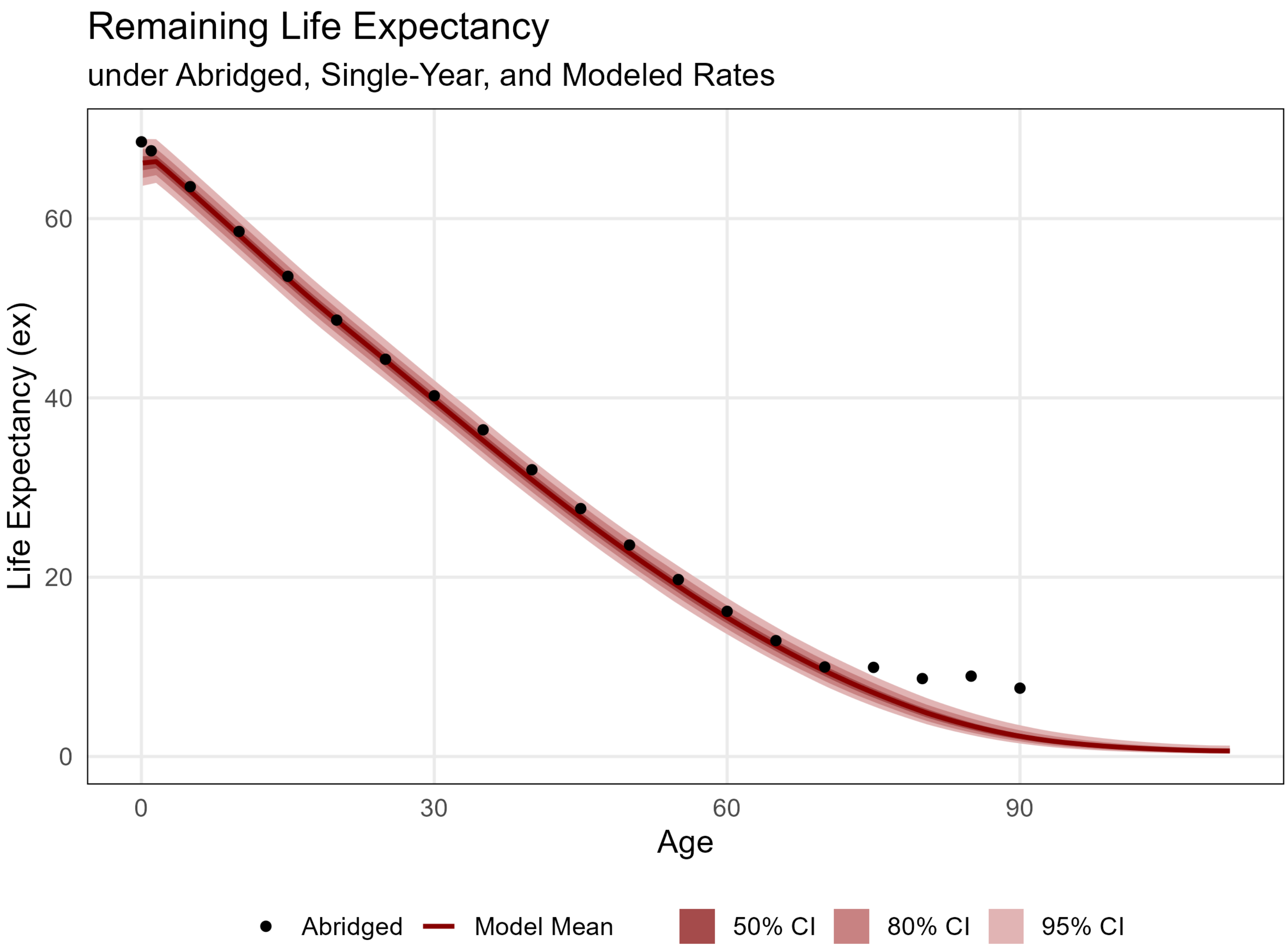
Scenario 2: Problematic Data

We create synthetic low-quality sparse data with:

- No reported deaths below age 15
- An implausible hump at ages 25-35
- Flat mortality above age 70
- Wide open-ended interval (90+)

We omit data for ages 0-5 and 80+ from the training data.

Age-Specific Mortality Rates
Abridged and Modeled
Log Mortality Rate (log mx)
0
-2
-4
-6
-8
0
30
60
90
Age
Abridged
Model Mean
50% CI
80% CI
95% CI



Remaining Life Expectancy
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Conclusion

1. **A General-Purpose Approach**: The proposed GARM approach is flexible, easy to implement with modern packages (`mgcv` in R), and works even with sparse/low-quality data.
2. **Structural Priors**: Combining a relational model with P-splines allows the model to "default" to demographically plausible patterns when data is sparse.
3. **Unified Framework**: Combines mortality smoothing, abridged to single-year conversion, and old-age extrapolation in a single model.
4. **Uncertainty Propagation**: Consistent uncertainty intervals for any life table quantity can be constructed via simulation, without the need of approximate procedures (e.g. via the delta method), or strong distributional assumptions (e.g. deaths counts with independent Poisson distributions).

Next Steps

1. **Extensive Testing**: We will test the model with different types of data, ideally with global coverage.

2. **Refinements**:

   - Provide guidance on how to choose a standard (when it matters).
   - Improve theory on selection of representative ages (a^*).

3. **Open-Source Code**: Release open source code to encourage use.

4. **Develop Educational Material**: Design a short course + an educational note or tutorial.